Robin Thibaut, PhD

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Summary

Internationally trained computational geoscientist who develops original **Bayesian frameworks** for uncertainty quantification and experimental design applied to geothermal, hydrological, and environmental systems. Creator of the open-source *SKBEL* library; published in premier journals including *Water Resources Research* (IF 4.6) and *Journal of Hydrology* (IF 6.4). Builds Python workflows that fuse finite-element thermal simulation, geophysics, and well data to **optimize drilling decisions** with **quantified risk**. Invited expert reviewer for 9 international journals (21 reviews, 2021–2024).

Core Skills

Programming & ML: Python (NumPy, Pandas, scikit-learn, PyTorch, TensorFlow); Uncertainty Quantification (Bayesian, BEL); Experimental Design; Physics-informed ML; Time-series & Spatial Modeling
Geoscience Modeling: Finite-element thermal simulation; MODFLOW, MT3DMS, MODPATH; ModelMuse; CRTOMO; RES2DINV; SGeMS; Geophysical Data Integration (ERT/IP); Hydrologic & Groundwater Modeling
Data & Engineering: SQL, Snowflake; Git; ETL/Data Engineering; Cloud (Google Cloud Platform); CI/CD; Scientific Visualization; Linux/macOS/Windows

Experience

Senior Computational Geoscientist – Zanskar Geothermal & Minerals, Salt Lake City, USA Oct 2024–Present
Geothermal energy startup developing next-generation exploration and resource assessment technology

- Architect and lead the geothermal modeling codebase: own internal repositories, enforce code review, unit tests, and CI/CD pipelines; manage releases ensuring reproducible modeling across the organization
- Prioritize drilling targets by fusing finite-element thermal simulations with **ML-based ranking**, directly reducing prospect uncertainty and raising decision confidence for multimillion-dollar drilling campaigns
- Design drilling target prioritization via **acquisition-function-based ranking** (success rate, expected improvement, upper confidence bound) over probabilistic subsurface temperature forecasts; deliver prioritized drilling recommendations with integrated techno-economic screening
- Calibrate 3D finite-element thermal models to well logs and geophysics; **history match** field observations and produce **P10/P50/P90** temperature and flow forecasts for executive decision support
- Engineer production data flows (Python, SQL/Snowflake, Git, GCP) with versioned ETL for site and simulation data, standardizing provenance and enabling dependable reruns across teams
- Deliver decision-ready maps and technical briefs to subsurface teams and executive leadership, streamlining cross-functional reviews and accelerating go/no-go recommendations

Postdoctoral Researcher – Lawrence Berkeley National Laboratory (LBNL), Berkeley, USA Aug 2023–Sep 2024
Premier U.S. Department of Energy national laboratory; Climate & Ecosystem Sciences Division, Watershed Function SFA

- Designed and implemented **spatiotemporal ML models** integrating geophysical, hydrologic, satellite, and field data to forecast watershed function under climate change scenarios
- Developed Python pipelines (Pandas/NumPy/scikit-learn/PyTorch) for automated feature engineering, cross-validation, and reproducible predictive modeling workflows
- Co-authored model–data integration study on snowmelt-driven mountain hydrology published in *Water Resources Research* (2025); led geophysical data processing and analysis components

Postdoctoral Researcher – Ghent University (TURBEAMS), Belgium Mar 2023–May 2023
BELSPO-funded national research program aboard the R/V Belgica

- Engineered the foundational deep-learning models linking multibeam water-column backscatter with turbidity/SPM for real-time water-quality monitoring
- Engineered scalable data pipelines handling millions of sonar data points per session, reducing model training wall-time

PhD Fellow – Ghent University, Belgium — Mar 2019–Mar 2023
Laboratory for Applied Geology and Hydrogeology

- Originated the **Bayesian Evidential Learning (BEL) experimental design** framework—a novel methodology that selects optimal measurements to minimize posterior uncertainty while reducing the number of required forward simulations
- Demonstrated that BEL-based geophysical monitoring designs outperform conventional well-only approaches for subsurface temperature characterization (*Water Resources Research* 2022), quantifying information-gain vs. cost trade-offs
- Published 3 first-author peer-reviewed articles in international journals and presented at 7 conferences; created and maintained open-source tools (*SKBEL*, *pysgems*) adopted by the research community

Project Engineer – G-tec, Belgium / International — Mar 2018–Jan 2019

- Executed offshore marine geophysical surveys (UXO detection, sub-bottom stratigraphy) across Poland, Scotland, and Vietnam with full QA/QC; delivered survey coverage on budget and schedule
- Coordinated field mobilization and HSE compliance, maintaining an exemplary safety record

Publications

Google Scholar: 177 total citations, h-index 6, i10-index 5 (as of March 2026)

Chen, Hang, **Robin Thibaut**, Chunwei Chou, Chen Xiong, and Yuxin Wu (2026). "A ModEx Framework for Watershed Subsurface Investigation with Limited Geophysical Data Using Machine Learning and Hydrologic Modeling". In: *Geophysical Research Letters* 53.2, e2025GL119953. doi: 10.1029/2025GL119953. Citations: 1.

Zhang, Le, Alexandros Daniilidis, Anne-Catherine Dieudonné, **Robin Thibaut**, and Thomas Hermans (2026). "A Bayesian Evidential Learning Framework for Safety and Performance Prediction in Thermo-Hydro-Mechanical Coupled Deep Mine Geothermal Systems". In: *Journal of Rock Mechanics and Geotechnical Engineering*. doi: 10.1016/j.jrmge.2025.11.022. Citations: 1.

Tas, Luka, Niels Hartog, Martin Bloemendal, David Simpson, Tanguy Robert, **Robin Thibaut**, Le Zhang, and Thomas Hermans (Jan. 2025). "Efficiency and heat transport processes of low-temperature aquifer thermal energy storage systems: new insights from global sensitivity analyses". In: *Geothermal Energy* 13. Open access, published 07 January 2025, Article number: 2. doi: 10.1186/s40517-024-00326-1. Citations: 9.

Wang, Lijing, Zexuan Xu, Chen Wang, **Robin Thibaut**, Craig Ulrich, Matthias Sprenger, Sebastian Uhlemann, Yuxin Wu, Evan King, Haruko Wainwright, Rosemary W. H. Carroll, Curtis Beutler, Kenneth H. Williams, and Baptiste Dafflon (2025). "The Role of Snowmelt and Subsurface Heterogeneity in Headwater Hydrology of a Mountainous Catchment in Colorado: A Model-Data Integration Approach". In: *Water Resources Research* 61.10, e2025WR040651. doi: 10.1029/2025WR040651. url: https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025WR040651. Citations: 2.

Zhang, Le, Anne-Catherine Dieudonné, Alexandros Daniilidis, Longjun Dong, Wenzhuo Cao, **Robin Thibaut**, Luka Tas, and Thomas Hermans (2025). "Thermo-Hydro-Mechanical Modeling of Geothermal Energy Systems in Deep Mines: Uncertainty Quantification and Design Optimization". In: *Applied Energy* 377, p. 124531. doi: 10.1016/j.apenergy.2024.124531. url: https://www.sciencedirect.com/science/article/pii/S0306261924019147. Citations: 34.

Thibaut, **Robin**, Nicolas Compaire, Nolwenn Lesparre, Maximilian Ramgraber, Eric Laloy, and Thomas Hermans (Nov. 2022). "Comparing Well and Geophysical Data for Temperature Monitoring Within a Bayesian Experimental Design Framework". In: *Water Resources Research* 58 (11). issn: 0043-1397. doi: 10.1029/2022WR033045. url: https://onlinelibrary.wiley.com/doi/10.1029/2022WR033045. Citations: 20.

Cong-Thi, Diep, Linh Pham Dieu, **Robin Thibaut**, Marieke Paepen, Huu Hieu Ho, Frédéric Nguyen, and Thomas Hermans (June 2021). "Imaging the Structure and the Saltwater Intrusion Extent of the Luy River Coastal Aquifer (Binh Thuan, Vietnam) Using Electrical Resistivity Tomography". In: *Water* 13 (13), p. 1743. ISSN: 2073-4441. DOI: 10.3390/w13131743. URL: https://www.mdpi.com/2073-4441/13/13/1743. Citations: 32.

Thibaut, **Robin**, Thomas Kremer, Annie Royen, Bun Kim Ngun, Frédéric Nguyen, and Thomas Hermans (Apr. 2021). "A new workflow to incorporate prior information in minimum gradient support (MGS) inversion of electrical resistivity and induced polarization data". In: *Journal of Applied Geophysics* 187, p. 104286. ISSN: 09269851. DOI: 10.1016/j.jappgeo.2021.104286. URL: https://linkinghub.elsevier.com/retrieve/pii/S0926985121000331. Citations: 22.

Thibaut, **Robin**, Eric Laloy, and Thomas Hermans (Dec. 2021). "A new framework for experimental design using Bayesian Evidential Learning: The case of wellhead protection area". In: *Journal of Hydrology* 603, p. 126903. ISSN: 00221694. DOI: 10.1016/j.jhydrol.2021.126903. URL: https://linkinghub.elsevier.com/retrieve/pii/S0022169421009537. Citations: 37.

Software and Datasets

Liang, Zhouji, Sofia Cecilia Brisson, **Robin Thibaut**, Junjie Yu, Carl Hoiland, and Ahinoam Pollack (2024). *Zanskar's GeoGym*. [**Dataset**]. URL: https://www.kaggle.com/datasets/robinthibautzanskar/zanskars-geogym.

Deleersnyder, Wouter and **Robin Thibaut** (2022). *Scale-dependent wavelet-based regularization scheme for geophysical 1D inversion*. [**Software**]. DOI: 10.5281/zenodo.6552695. URL: http://dx.doi.org/10.5281/zenodo.6552695.

Lesparre, Nolwenn, Nicolas Compaire, Thomas Hermans, and **Robin Thibaut** (2022). *4D Temperature Monitoring*. [**Dataset**]. DOI: 10.34740/kaggle/dsv/3819983. URL: https://www.kaggle.com/dsv/3819983.

Thibaut, **Robin** (2021). *WHPA Prediction*. [**Dataset**]. DOI: 10.34740/kaggle/dsv/2648718. URL: https://www.kaggle.com/dsv/2648718.

Thibaut, **Robin** and Maximilian Ramgraber (Sept. 2021). *SKBEL - Bayesian Evidential Learning framework built on top of scikit-learn*. [**Software**]. Version v2.0.0. DOI: 10.5281/zenodo.6205242. URL: https://doi.org/10.5281/zenodo.6205242. Citations: 4.

Thibaut, **Robin** and Guillaume Vandekerckhove (May 2021). *pysgems–Use SGeMS (Stanford Geostatistical Modeling Software) within Python*. [**Software**]. Version v1.3. DOI: 10.5281/zenodo.4773587. URL: https://doi.org/10.5281/zenodo.4773587.

Invited Expert Peer Review

- Invited by editorial boards to review 21 manuscripts across 9 internationally ranked journals (2021–2024): *Water Resources Research* (IF 4.6), *Geophysical Research Letters* (IF 5.2), *Journal of Hydrology* (IF 6.4), *Advances in Water Resources* (IF 4.7), *Geophysical Journal International* (IF 2.8), *GEOPHYSICS* (IF 3.3), *Pure and Applied Geophysics* (IF 1.9), *Hydrogeology Journal* (IF 2.4), *GEUS Bulletin*

Selected Public Projects

Rare Earth Elements — Multiphysics AI-aided Autonomous Prospecting (REE-MAP) Pennsylvania, USA (2023–ongoing)

- Led geophysical data collection, processing, inversion, and interpretation in a multiphysics, AI-aided approach to identify REE–CM "hot zones" in coal tailings; field-tested at multiple sites; GSA Connects 2024 abstract with authorship credit and DOI.
  GSA Abstract, Paper 404345, DOI: 10.1130/abs/2024AM-404345

TURBEAMS — Towards 3D turbidity by correlating multibeam sonar and in-situ sensor data Belgium (project 2021–2026; contributor 2023)

- Developed deep-learning models linking multibeam water-column backscatter with turbidity/SPM; delivered Python data flows enabling **3D turbidity/SPM imaging** plus a new processing library and workflows.
  National research program (New RV Belgica, BELSPO) with published ML objectives (incl. BEL) for large

multibeam/optical datasets.

Imaging Saltwater Intrusion, Luy River Coastal Aquifer (Vietnam) Vietnam (2019–ongoing)

- Co-authored an open-access ERT study delineating the saline boundary; **21 ERT profiles** revealed a significantly larger intrusion zone than previously recognized, informing regional water-resource management.
  Water 2021 (MDPI), DOI: 10.3390/w13131743

Bayesian Evidential Learning (BEL) for Experimental Design in Earth Sciences Belgium (2019–2023)

- Originated **BEL-based experimental design** methodology that minimizes posterior uncertainty and optimizes monitoring network design; resulted in peer-reviewed publications and the **SKBEL** open-source package adopted internationally.
  Journal of Hydrology 2021, 10.1016/j.jhydrol.2021.126903; *Water Resources Research* 2022, 10.1029/2022WR033045

Education

Ph.D. in Geological Sciences – Ghent University 2019–2023
Laboratory for Applied Geology and Hydrogeology
M.Sc. in Geological Engineering – University of Liège 2014–2017
Cum Laude
B.Sc. in Geological Sciences – Free University of Brussels 2010–2014
Cum Laude

Conferences and Talks

Liang, Zhouji, Junjie Yu, **Robin Thibaut**, Fangning Zheng, Carl Hoiland, and Ahinoam Pollack (Feb. 2026). "Surrogate Modeling for Geothermal Systems: Accelerating Optimization, History Matching, and Uncertainty Quantification". In: *Proceedings, 51st Workshop on Geothermal Reservoir Engineering*. SGP-TR-230. Stanford, California, pp. 1–10.

Zheng, Fangning, **Robin Thibaut**, Zhouji Liang, Junjie Yu, and Ahinoam Pollack (Feb. 2026). "Thermal Gradient Signatures of Fault-Controlled Hydrothermal Systems: Insights from Conceptual Models and Field Data". In: *Proceedings, 51st Workshop on Geothermal Reservoir Engineering*. SGP-TR-230. Stanford, California, pp. 1–8.

Liang, Zhouji, Sofia Cecilia Brisson, **Robin Thibaut**, Junjie Yu, Carl Hoiland, and Ahinoam Pollack (Feb. 2025). "GeoGym: a Benchmark Dataset for Evaluating Geothermal Exploration Strategies". In: *Proceedings, 50th Workshop on Geothermal Reservoir Engineering*. SGP-TR-229. Stanford, California, pp. 1–7. URL: https://pangea.stanford.edu/ERE/db/GeoConf/papers/SGW/2025/Liang.pdf.

Wu, Yuxin, Karthika Balan, Chunwei Chou, Jun Woo Chung, Baptiste Dafflon, Nicola Falco, Jim Panaro, Arun Persaud, Brian Quiter, Emil Rofors, **Robin Thibaut**, Jiannan Wang, Michael L. Whittaker, and John Wu (2024). "RARE EARTH ELEMENTS – MULTIPHYSICS AI-AIDED AUTONOMOUS PROSPECTING (REE -MAP)". In: *Geological Society of America Abstracts with Programs, GSA Connects 2024 Meeting in Anaheim, California*. Vol. 56. 5. Anaheim, U.S.A., p. 1. DOI: 10.1130/abs/2024AM-404345.

Thibaut, **Robin**, Ty Ferré, and Thomas Hermans (2023). "Sequential Optimization Of Temperature Measurements To Estimate Groundwater Surface Water Interactions". In: *AGU Fall Meeting 2023, Abstracts*. San Francisco, U.S.A., p. 1.

Thibaut, **Robin**, Nicolas Compaire, Nolwenn Lesparre, Maximilian Ramgraber, Eric Laloy, and Thomas Hermans (2022). "Comparing well and geophysical data for temperature monitoring within a Bayesian Experimental Design framework". In: *Computational Methods in Water Resources, Abstracts*. Gdansk, Poland, p. 20.

Hermans, Thomas, Nicolas Compaire, **Robin Thibaut**, and Nolwenn Lesparre (2021). "Bayesian evidential learning : an alternative to hydrogeophysical coupled inversion". In: *First International Meeting for Applied Geoscience & Energy Expanded Abstracts*. online, pp. 3125–3129. URL: http://dx.doi.org/10.1190/segam2021-3580979.1.

Thibaut, **Robin**, Thomas Hermans, and Eric Laloy (2021a). "A new framework for experimental design using Bayesian Evidential Learning : the case of wellhead protection area". In: *AGU Fall Meeting 2021, Abstracts*. New Orleans, U.S.A., p. 1.

Thibaut, **Robin**, Thomas Hermans, and Eric Laloy (2021b). "Bayesian Evidential Learning combined with experimental design : the case of wellhead protection area prediction". In: *48th IAH Congress, Abstracts*. Brussels, Belgium. URL: https://iah2021belgium.org/wp-content/uploads/2021/09/IAH-2021-Book-of-Abstracts.pdf.

– (2020). "A new framework to reduce uncertainty in Wellhead Protection Area prediction using Bayesian Evidential Learning". In: *10th International Congress on Environmental Modelling and Software (iEMSs 2020), Abstracts*. Online, p. 1. URL: https://iemss2020.com/.

Thibaut, **Robin**, Thomas Kremer, Annie Royen, Bun Kim Ngun, Frederic Nguyen, and Thomas Hermans (2019). "A new approach to incorporate prior information in MGS inversion of ERT/IP data". In: *Near Surface Geoscience conference and exhibition 2019*. The Hague, The Netherlands: EAGE, p. 5. URL: http://dx.doi.org/10.3997/2214-4609.201902391.

Teaching

Teaching Assistant – Groundwater Modeling course, Ghent University 2019–2023
Organizer and Speaker – Python Workshop, Ghent University Mar 2022
Private Tutor – Mathematics and Physics for university students 2016–2019

Languages

English (Fluent), French (Fluent), Dutch (Intermediate), Vietnamese (Intermediate)